Where a Quantum Reservoir Works: A Transferable Operating Band

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***Abstract*—Quantum reservoir computing performs temporal machine learning by training only a linear readout on the measured outputs of a fixed quantum system, leaving the quantum dynamics itself untrained. This makes the approach attractive for near-term hardware, but it sharpens a question that has remained largely unresolved: which device settings render the fixed dynamics useful rather than merely high-dimensional. We address this question for a stateful dissipative gate-model reservoir and show that strong performance concentrates in a well-defined operating region governed by input drive, recurrent coupling, and controlled dissipation. This region is transferable: an operating band identified on a subset of tasks predicts effective settings on held-out tasks and held-out reservoir initializations alike. Mechanism ablations establish that the region is causal rather than incidental, and an inexpensive memory-based diagnostic recovers it prior to task-specific evaluation.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes**

I. Introduction

Reservoir computing rests on an unusually economical idea: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional trace of its recent history, and learning reduces to fitting a single linear readout on top of that trace [1]–[4]. The dynamics are never optimized, only observed. This economy is precisely what makes the approach appealing for near-term quantum hardware: native quantum evolution can serve as the reservoir and only the classical readout must be trained [5]–[8]. Yet the same economy relocates the entire burden of the method onto the fixed dynamics, and so onto a question that is easy to state and surprisingly hard to answer: what makes those dynamics useful?

Recent QRC work now spans noisy gate-model devices, continuous-variable and oscillator reservoirs, weak-measurement protocols, photonic reservoirs, noise-engineered reservoirs, and experimental optical memory control [9]–[14]. Classical reservoir theory has long since refused the obvious answer. Usefulness is not maximal expressivity. A reservoir earns its keep by holding a balance—retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [4], [15]. A quantum reservoir inherits this lesson but complicates it, because the three ingredients are no longer free dials. They are downstream consequences of distinct physical resources: how strongly the input perturbs the carried state, how coherently the system mixes that state across its parts, and how it is allowed to dissipate [7], [16], [17]. A large Hilbert space supplies dimension, but dimension alone preserves no memory; drive that is too strong overwrites history, and dynamics with no dissipation never forget in the way fading memory requires.

We take this seriously as an organizing principle rather than a caveat. For a stateful dissipative gate-model reservoir, we treat the space of these three resources as the proper setting in which to ask when the reservoir works, and we map it. The picture that emerges is not a single tuned operating point but a connected region—gentle input, nonmaximal coupling, controlled damping—in which good performance concentrates and from which it does not stray arbitrarily across tasks. The contribution of this paper is that region itself: its existence, its transfer across tasks and reservoir initializations, the mechanisms that sustain it, and a memory-based diagnostic that locates it before any task-specific search.

II. Protocol

A. *Reservoir*

The reservoir processes a scalar time series without resetting its density matrix, so the carried state stores recent history. At step t, the new input perturbs the previous state through a uniform rotation,

$$\rho_{t,0} = U_{\rm in}(u_t)\rho_{t-1,L}U_{\rm in}^{\dagger}(u_t), \qquad U_{\rm in}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

The input strength β controls how strongly each step writes into the carried state.

The perturbed state then passes through identical layers: a frozen local mixer, ring ZZ coupling, and a dissipative channel,

$$\begin{aligned} \rho_{t,\ell} &= \mathcal{D}_\gamma^{\otimes n}\left[U_\lambda U_{\rm mix}\rho_{t,\ell-1}U_{\rm mix}^{\dagger}U_\lambda^{\dagger}\right], \\ U_\lambda &= \prod_{(i,j)\in E_{\rm ring}} e^{-i\lambda Z_i Z_j}. \end{aligned} \tag{2}$$

The coupling strength λ controls recurrent mixing and γ controls forgetting. We vary only $\theta = (\beta, \lambda, \gamma)$; all other gates are frozen and only a linear readout is trained. The three axes therefore correspond directly to input drive, nonlinear mixing, and controlled forgetting.

This architecture is used as a minimal testbed, not as an optimized quantum model. Four qubits and two layers keep the full phase grid exhaustively searchable, while the ring coupling still produces recurrent many-body mixing. The $Z{+}ZZ$ readout is deliberately modest: it observes local and nearest-neighbor correlations rather than the full density matrix. This makes the

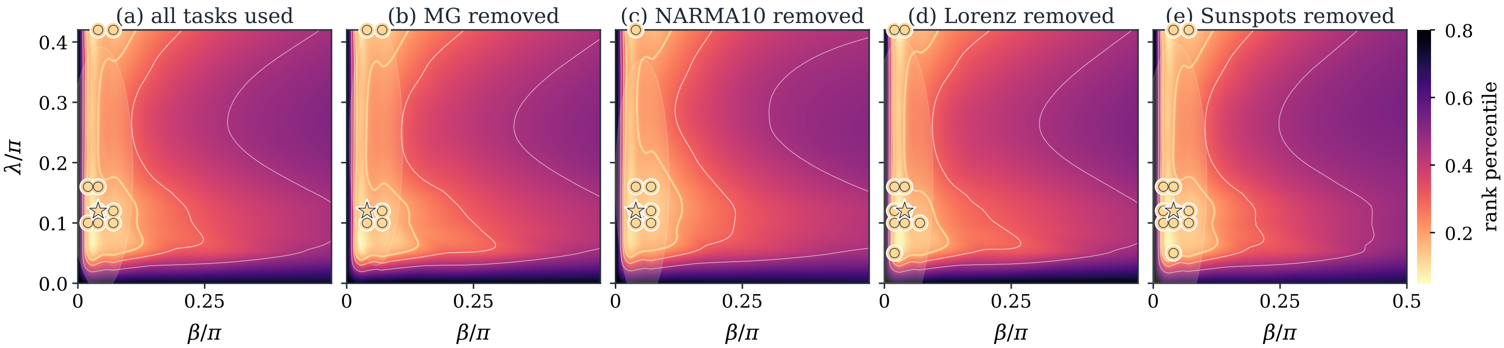

Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. The first panel uses all four tasks. Each following panel recomputes the map after removing one task from the validation aggregate. If the bright low-rank region remains in the same place, the operating band is not driven by a single task. Gold circles mark coordinates that repeatedly fall in the top validation quintile; the star marks the primary-band medoid projected onto this slice.

phase maps interpretable as operating-regime evidence instead of architectural tuning.

B. Configuration

Unless stated otherwise, we use four qubits, two layers, ring coupling, uniform input injection, amplitude damping, and a readout from single-qubit Z and ring-pair ZZ expectations. Training is ridge regression on these features alone. The grid spans $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, and $\gamma = \{0, 0.01, 0.05, 0.12, 0.22, 0.30\}$ over seeds 42–61. Hyperparameters are selected on validation data only; holdout data never define the operating region. The resulting balanced panel contains $4 \times 20 = 80$ task-seed replicates for every grid coordinate.

C. Tasks, Metric, and Splits

We use four scalar one-step prediction tasks: Mackey–Glass chaos [18], the next x coordinate of an RK4 Lorenz trajectory [19], NARMA10 memory prediction [20], and annual Sunspots from `statsmodels`. Mackey–Glass, Lorenz, and NARMA10 use washout/train/validation/holdout lengths 300/1600/400/400; Sunspots uses 20/150/50/60. Inputs are scaled to $[-1, 1]$ and targets are standardized. The error is normalized mean-squared error, $\mathrm{NMSE} = \langle (y - \widehat{y})^2 \rangle / (\mathrm{Var}(y) + 10^{-12})$, computed separately on validation and holdout splits. Ridge is selected only by validation NMSE from $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$.

D. Operating Band

To locate the useful region we rank performance within each task j and seed s. A setting θ receives validation percentile rank $r_{j,s}(\theta)$, with lower rank better. The operating band contains settings that land in the top fraction p for at least a fraction q of task-seed pairs,

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

This frequency-level definition asks whether a region behaves well repeatedly, not whether one coordinate wins every task. In leave-one-task or leave-one-seed tests, the band is built from the remaining replicates and scored on the withheld replicate. In the holdout audit, the validation-defined band is frozen first and evaluated once on holdout ranks and NMSE. Confidence intervals are nonparametric bootstrap percentile intervals for the mean with 50,000 resamples and a fixed seed. Because ranks are formed within each task-seed replicate before aggregation, large-error tasks cannot dominate small-error tasks and difficult seeds cannot dominate easier seeds. The band is therefore a repeated-success criterion over the full panel, not an average-error threshold.

III. Results

Across the control space, low validation rank does not spread arbitrarily; it gathers into a compact, connected region, and relaxing the selection threshold grows that region smoothly rather than fragmenting it—the signature of a single operating regime rather than a scatter of coincidences (Fig. 1). The damping coordinate is selected by the validation scan, not fixed by hand: the near-zero $\gamma = 0.01$ slice is a weak contrast, while the strict $B_{20,0.7}$ core appears in the nonzero damping slices $\gamma = 0.12$–0.30 with medoid $\gamma = 0.22$ (Fig. 2). A region that merely fit the four tasks jointly would prove little, so we test whether it survives the removal of the information it was built from. It does: when one task is withheld and the band is defined from the rest, the held-out task still lands in the top quartile on average, with mean validation rank 0.193 and CI $[0.176, 0.210]$; when a seed is withheld instead, the held-out seed behaves the same way, with mean rank 0.151 and CI $[0.139, 0.163]$. The region is thus a search prior that generalizes across both task identity and reservoir initialization, not a fit to either.

The same validation-defined band also holds on data it never touched: after the band is frozen, evaluating once on the chronological holdout split leaves it top-quartile on unseen data (Table I). Repeating the identical protocol on a nearby four-qubit, three-layer variant tests whether the phenomenon is tied to the exact depth choice; the table reports band existence, held-out rank, and Jaccard overlap with the base control

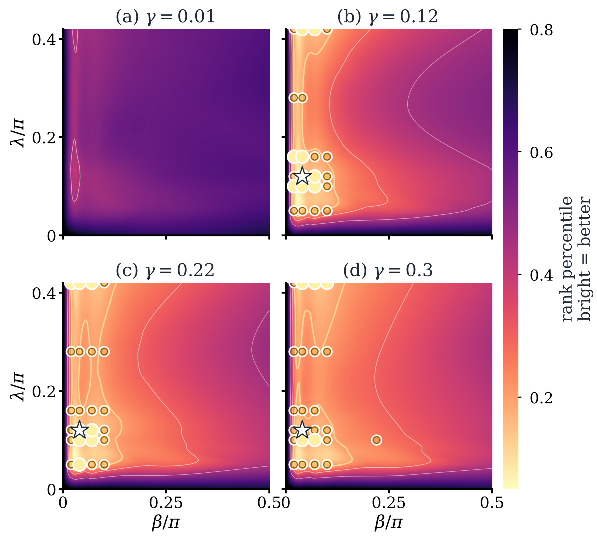

Fig. 2. Damping-slice check for the same validation-ranked grid. Bright color indicates better validation rank. The near-zero damping slice at $\gamma = 0.01$ provides a contrast, while the operating band appears across stronger nonzero damping slices.

TABLE I
HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. MEANS ARE OVER TWENTY SEEDS PER TASK; "ALL" AVERAGES THE 80 TASK-SEED ROWS.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	7.05e-05	5.26e-05	0.104
Lorenz	1.51e-05	1.52e-05	0.136
NARMA10	0.560	0.607	0.204
Sunspots	0.209	0.209	0.168
All	0.192	0.204	0.153

region (Table II). That the band is real rather than cosmetic is confirmed by perturbing the dynamics that sustain it: removing damping, removing recurrent mixing, or replacing amplitude damping with dephasing each erases the region entirely, with retention values 0.00, 0.00, and 0.00 under the same criterion. The collapse is total rather than gradual, exactly as expected if the regime is held in place by input strong enough to write but not overwrite, mixing that builds nonlinear features, and damping that preserves usable memory rather than merely adding noise (Fig. 3). Finally, the region can be found before any task-specific search: memory capacity correlates strongly and negatively with validation rank, $\rho_s = -0.88$, as do the information-processing capacities [15], whereas raw feature diversity does not (Fig. 4).

IV. Discussion and Scope

For this reservoir family, useful learning is organized by a transferable operating regime rather than by task-specific tuning—a hypothesis whose credibility comes from variety. Four tasks with little in common, spanning smooth chaos, nonlinear memory, and irregular real-world data, obey the same region, and that region holds when any one task or any one seed is withheld; what survives this much variation is hard to dismiss as a fluke. The reading we draw is deflationary.

TABLE II
ARCHITECTURE ROBUSTNESS AFTER VALIDATION-ONLY BAND SELECTION. OVERLAP IS JACCARD OVERLAP WITH THE BASE BAND IN THE SHARED (β, λ, γ) GRID.

Setting	Band?	Held-out rank	Overlap
4q, 2 layers, Z+ZZ	yes	0.153	1.00
4q, 3 layers, Z+ZZ	yes	0.154	0.04

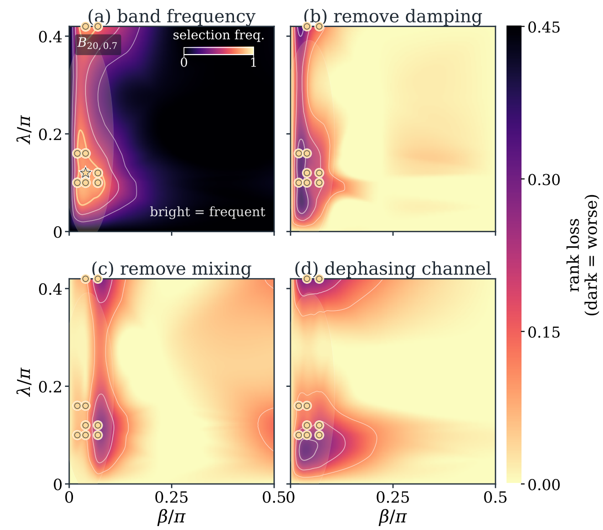

Fig. 3. Why the operating band is meaningful. (a) Bright coordinates are frequently selected by validation. (b–d) Removing damping, removing mixing, or replacing amplitude damping by dephasing concentrates rank loss around the same gold band, showing that the useful region depends on these mechanisms.

The useful region is not picked out by Hilbert-space size or by feature diversity but by memory, which suggests that a quantum reservoir succeeds not because its state space is large but because drive, coupling, and dissipation together keep memory and nonlinearity in balance—the principle classical reservoir theory named long ago [4], [15]. This is also why the ablations bite: strip out damping or mixing and the regime does not weaken, it disappears. The boundaries are drawn on purpose. Everything here is simulated, on one architecture family, with measurement treated as free, so finite-shot noise, calibration drift, and readout cost are all absent and could each deform the regime [11], [21]; we therefore claim neither quantum advantage nor hardware efficiency. The claim is smaller and sturdier—that the operating regime is the right object to study, that it transfers, and that memory is how to find it.

V. Conclusion

The usefulness of a dissipative quantum reservoir is not a property of size alone but of where it is operated. Across tasks and reservoir initializations, strong performance gathers in a transferable region where input drive, recurrent coupling, and dissipation hold memory and nonlinearity in balance. The regime transfers, collapses under targeted mechanism removals, and can be found in advance by memory-based diagnostics rather than raw feature diversity. The result is intentionally limited: all experiments are simulated, measurement is treated

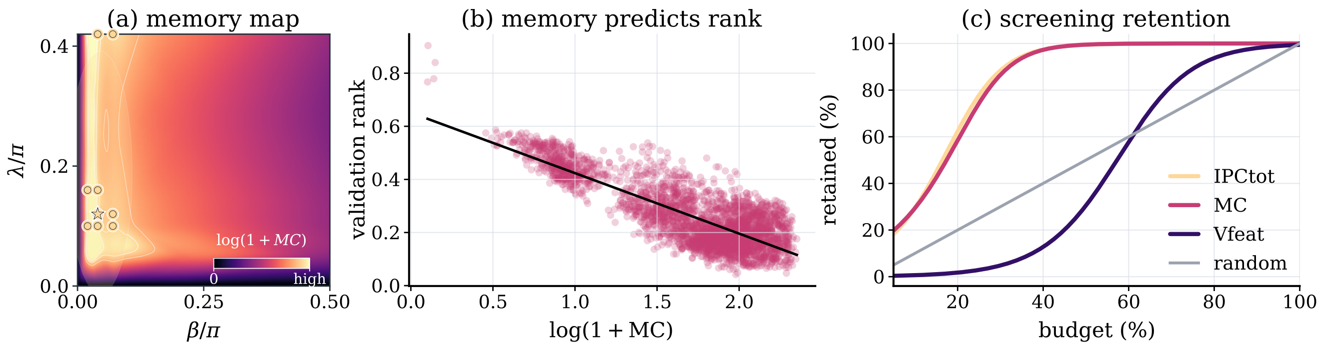

Fig. 4. Memory-based diagnostics identify the operating region before task-specific search. (a) The task-free memory map shows $\log(1+\mathrm{MC})$ on the same $\gamma = 0.12$ control slice as the phase maps. The gold operating-band points lie inside the structured memory-rich region, showing that the band is visible from reservoir dynamics alone. (b) Across all sampled settings and seeds, higher memory predicts lower validation rank, giving a global quantitative version of the same effect. (c) Screening efficiency: the fraction of genuinely good reservoirs retained as a function of how many settings are kept. Screening by memory or information-processing capacity recovers the useful settings far earlier than random selection, while raw feature diversity remains much weaker.

as free, and finite shots, calibration drift, device noise, and readout cost remain outside the present claim.

Reproducibility appendix.

Package. The project repository contains the code, checked-in CSV/JSON results, and final figures used in this paper.

Frozen reservoir. The simulator evolves the full density matrix exactly, with no finite-shot sampling. It starts from $\rho_0 = |0\cdots0\rangle\langle 0\cdots 0|$ and is carried between time steps without reset. For each seed, `default_rng(s)` draws one fixed mixer $U_{\mathrm{mix}} = \bigotimes_i R_x(\theta_i)$ with $\theta_i \sim U[0, 2\pi)$; this mixer is reused at every step and layer. Each layer applies $U_\lambda U_{\mathrm{mix}}$ on ring edges, then local amplitude damping. The readout is ridge regression on exact Z_i and ring-pair $Z_i Z_j$ expectations, with an unpenalized intercept.

Selection and diagnostics. The (β, λ, γ) grid and task splits are in Sec. II. Ridge penalties $\{10^{-8}, \ldots, 10^3\}$ are chosen by validation NMSE only; holdout data are used only after the band is fixed. MC/IPC diagnostics use an independent $U[-1, 1]$ drive of length 900, washout 150, ridge $\alpha = 10^{-8}$, and a 70/30 chronological split. MC sums held-out positive R^2 over delays 1–10. IPC adds linear delays 1–20, quadratic Legendre delays 1–10, and ten cross-delay products, clipping negative R^2 to zero. CIs are nonparametric bootstrap percentile intervals with 50,000 resamples and a fixed seed.

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